



RAJARATA UNIVERSITY OF SRI LANKA

FACULTY OF APPLIED SCIENCES

Department of Computing

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Group 10: Frame Enhancement Method (Submission – Phase 03)

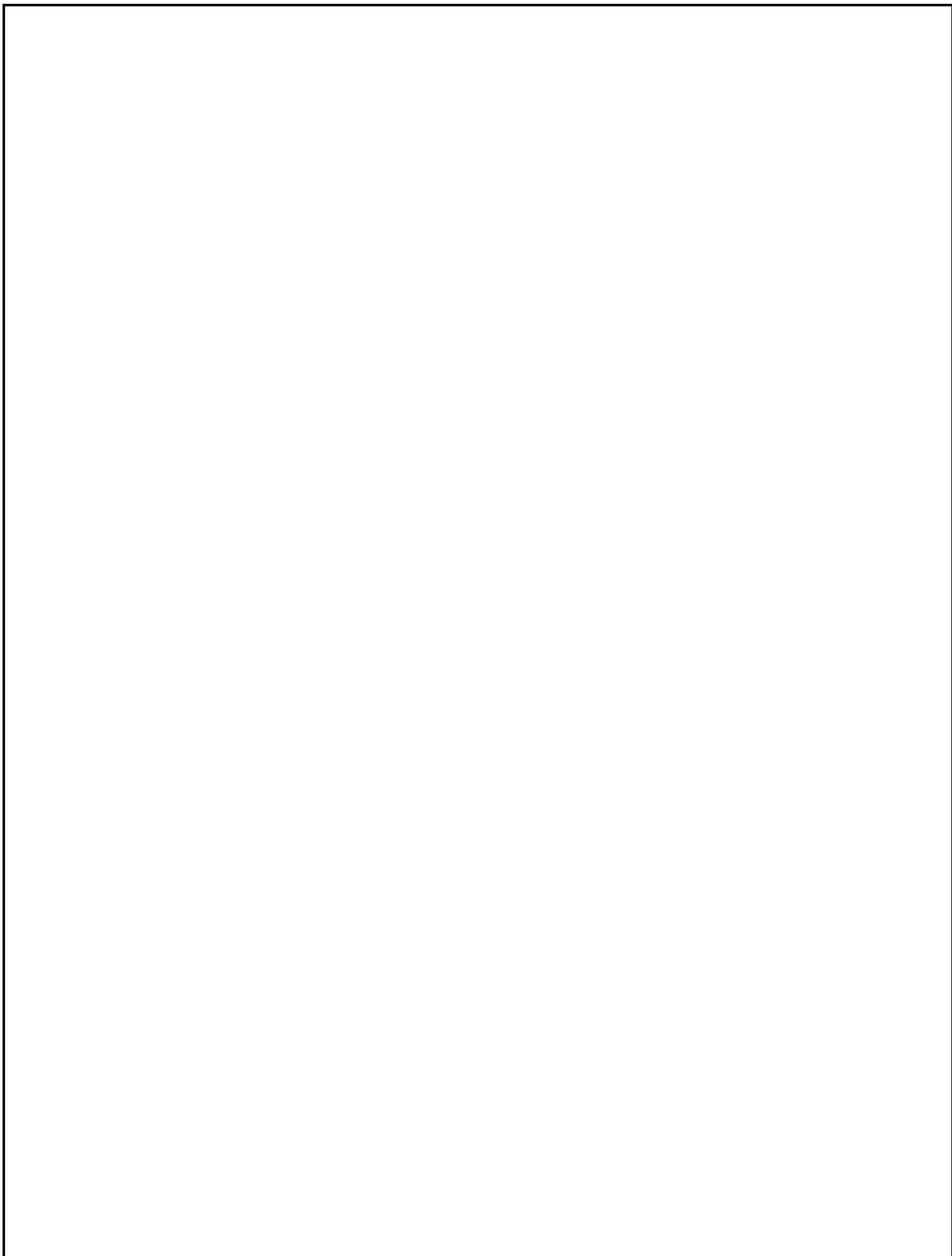
AUTOMATED ROAD CONDITION MONITORING SYSTEM

Submitted By:

Name	Reg. No	Index No.	Email
U P N M Pathirana	ICT/2023/046	6036	ict2023046@as.rjt.ac.lk
W D P P Witharana	ICT/2023/045	6035	ict2023045@as.rjt.ac.lk
H P D N D Ariyathilaka	ICT/2023/044	6034	ict2023044@as.rjt.ac.lk
J G A D Madhushani	ICT/2023/043	6033	ict2023043@as.rjt.ac.lk

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01.Introduction to Frame Enhancement

The Automated Road Condition Monitoring System is a computer-vision-based project designed to identify and analyze road damages like potholes, alligator cracks, and raveling. In real-world scenarios, videos recorded from moving vehicles often suffer from quality issues like graininess, blur, and bad lighting. These problems make it difficult for AI models to "see" the damage clearly. Therefore, a specialized Enhancement Pipeline is required to clean and prepare these frames before they can be used for final analysis.

Most raw video frames captured on the road are affected by environmental factors. Sensor noise (grain) and compression artifacts can distort small details, while uneven illumination and tree shadows create dark spots that hide cracks. Additionally, motion blur caused by vehicle vibrations can soften the edges of potholes, making them look like blurry blobs. These degradations lower the accuracy of the system, making it hard to distinguish between a simple shadow and a dangerous road defect.

To fix these issues, our project uses a step-by-step Image Enhancement Pipeline. We first extract the frames and convert them to Grayscale to focus on the texture of the road. We then use Adaptive Denoising, where the system automatically chooses the best filter (Median or Geometric Mean) based on how much noise is detected. To fix lighting, we apply CLAHE (Contrast Limited Adaptive Histogram Equalization) and Contrast Stretching, which balances the shadows and highlights so the road damage becomes clearly visible.

The final part of the pipeline focuses on Sharpness Evolution, restoring the crisp edges of cracks that were lost during recording. By moving the image through these sequential stages, we gradually remove artifacts and highlight the most important features. These enhanced frames provide the high quality and edge detail needed for the final Segmentation and Damage Detection stages, ensuring the system is reliable even in difficult real-world conditions.

02.Identification of Frame Degradations

To identify the specific quality issues impacting image clarity, a comprehensive analysis of the extracted frames was conducted. We utilized a combination of Visual Examination, Global Histogram Observation, and Laplacian Variance Scoring. This multi-method approach allowed us to quantify the severity of degradations across different road conditions, such as potholes, alligator cracks, and raveling.

2.1. Noise Analysis

We used the Laplacian Variance operator to compute a quantitative Noise Score for each grayscale frame. This allowed us to categorize noise into three distinct levels based on the intensity of pixel fluctuations:

- **Sensor Noise (Score < 300):** Appearing as fine, pseudo-random grain caused by the CMOS imaging sensors, especially visible in shaded road segments.
- **Compression Noise (Score 300–900):** Resulting from the MP4 (H.264) video encoding. This appears as "block artifacts" at road texture boundaries, which reduces the discriminability of thin cracks.
- **Extreme Noise (Score > 900):** Found in low-light pothole categories where sensor grain and high-frequency compression artifacts combine.

Impact on ROI Detection: Noise introduces "false edges." During the segmentation stage, the system might incorrectly interpret these random pixel variations as small cracks, leading to false-positive detections.

ORIGINAL: grayscale_alligator_Cracking01 (Frame 170)

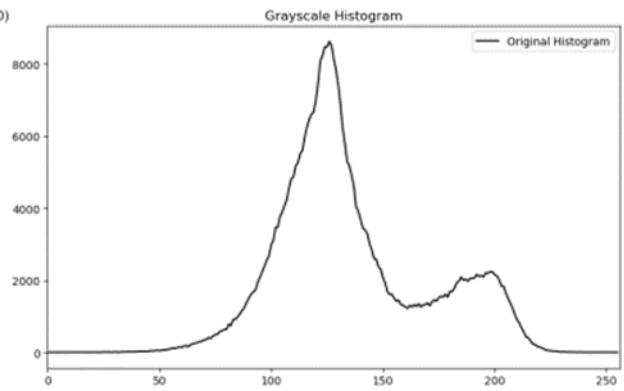
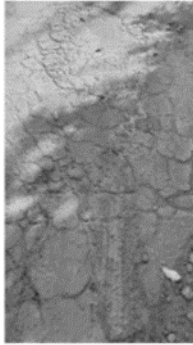


Figure 1: Grayscale image and its corresponding intensity histogram- Alligator Cracking

DENOISED (Median Filter)

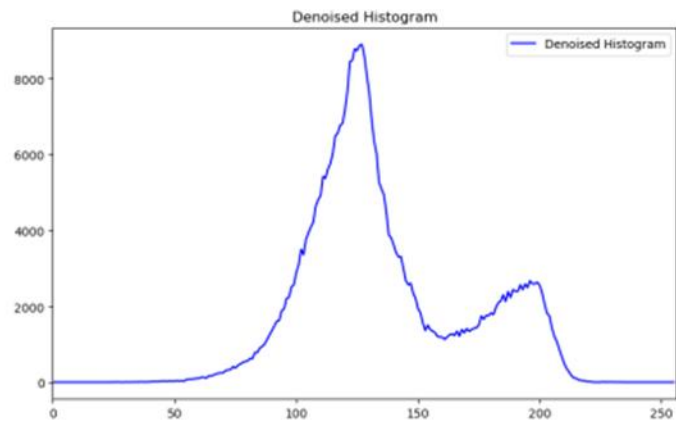
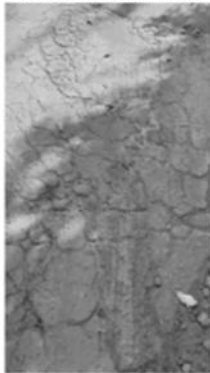


Figure 2: Denoised image and its corresponding histogram- Alligator Cracking

DENOISED (Hybrid: Median + Geometric)

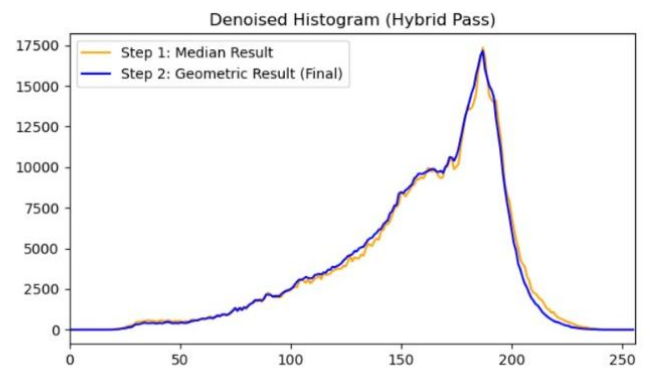
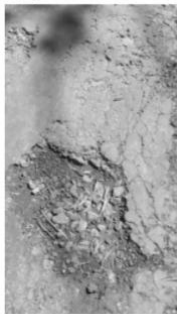


Figure 3: Hybrid Denoising Analysis (Median + Geometric Mean) - Pothole

2.2. Illumination Variations and Shadows

By analyzing the pixel intensity histograms, we identified significant narrow dynamic ranges. The histograms revealed two main issues:

- **Overexposure:** High pixel density near the 200–255 range, indicating "blown-out" road surfaces where detail is lost in bright sunlight.
- **Underexposure/Shadows:** Large peaks in the 0–50 range, caused by tree shadows or recording equipment. These dark patches hide the actual depth of potholes and the length of cracks.

Impact on ROI Detection: Uneven lighting reduces the contrast between the damaged region (darker) and the undamaged pavement. If the histogram is too "bunched up" in one area, the segmentation algorithm cannot find a clear threshold to separate the crack from the road.

Grayscale Image (Frame 170)

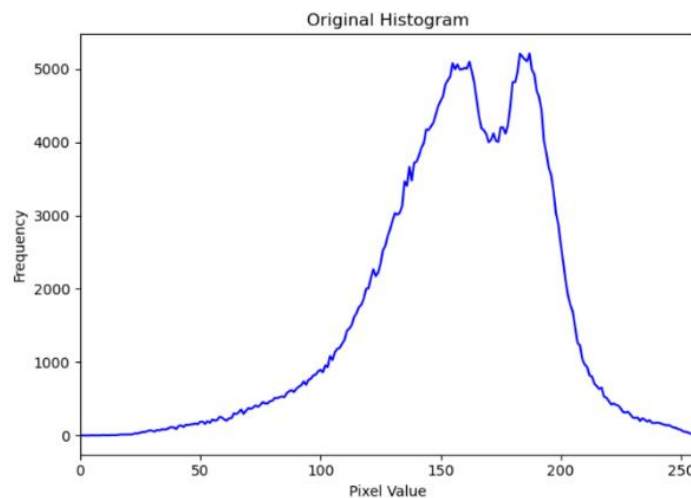


Figure: Intensity histogram showing underexposure and narrow dynamic range caused by environmental shadows.

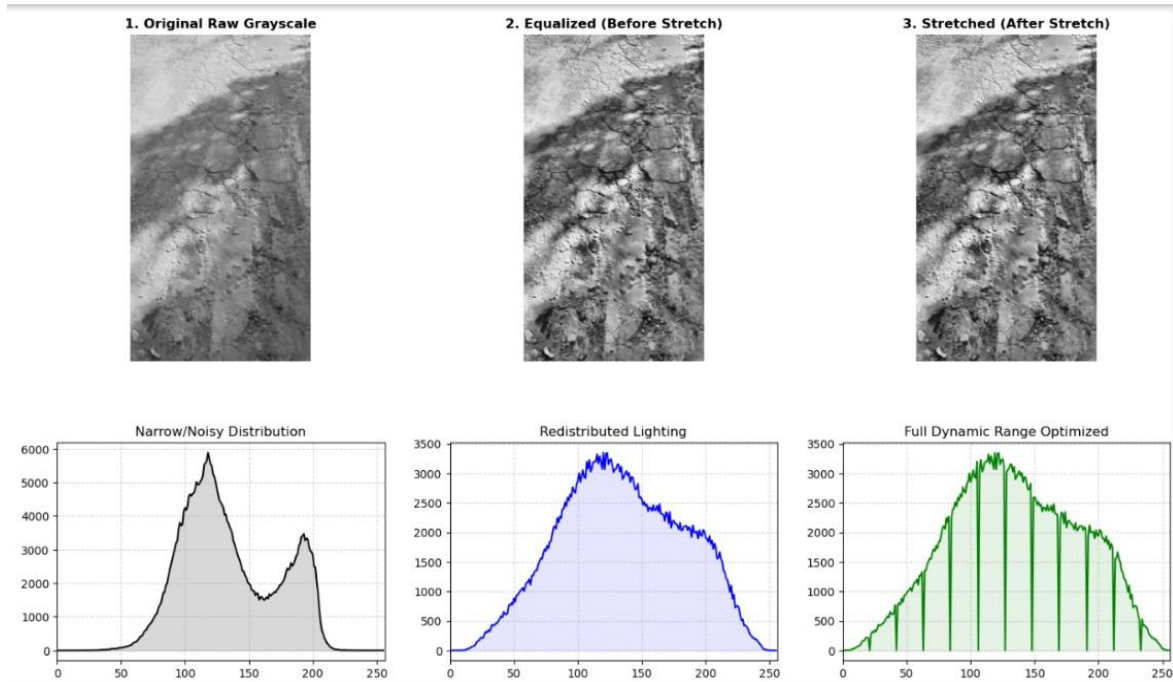


Figure: Grayscale image, histogram equalized image, and contrast stretched image with their corresponding histograms

2.3. Motion Blur and Camera Instability

Since the videos were recorded from a moving vehicle, handheld instability and engine vibrations introduced **Motion Blur**. We detected this by calculating the **Sharpness Evolution Score** (using the variance of the Laplacian).

- **Observation:** Blurred frames show "softened" edges. When we apply a Canny Edge detector to these frames, the lines for cracks appear broken or fuzzy rather than sharp and continuous.

Impact on ROI Detection: Blur destroys the high-frequency details needed to locate exact damage boundaries. This makes it difficult to accurately measure the width of a crack or the precise border of a pothole.

2.4. Summary of Detected Degradations

The analysis confirms that raw frames are unsuitable for direct segmentation. The combination of high-frequency noise, inconsistent lighting, and vibration-induced blur creates a "noisy" environment for AI detection.

Degradation Type	Detection Method	Visual Evidence
Noise	Laplacian Variance	Image looks grainy or pixelated
Illumination	Histogram Observation	Spikes in dark or bright regions
Blur	Gradient Magnitude	Edges are unclear or blurry
Contrast	Standard Deviation	Pixel values are crowded together in the histogram

To minimize these issues, we have designed the following **Enhancement Pipeline** to normalize the frames before performing any damage detection.

03. Pre – Processing of Video Frames

The initial phase of the pipeline focuses on preparing the raw data for analysis. This involves converting high-volume video data into manageable image files and optimizing their color properties for texture analysis.

3.1. Step 1: Frame Extraction

The dataset consists of nine MP4 video recordings covering various road conditions. To avoid data redundancy and ensure efficient processing, we used the cv2.VideoCapture library in Python to extract frames at specific intervals.

Video Category	Damage Type	Sampling Rate
alligatorCracking01	Alligator Cracking	Every 10th frame
transversCracking01	Transverse Cracking	Every 10th frame
crossing_Marking_Removal02	Crossing Marking Removal	Every 5th frame
raveling01	Raveling	Every 20th frame
lane_Marking_Removal01	Lane Marking Removal	Every 10th frame
potHoles01	Potholes Type A	Every 10th frame
potHoles02	Potholes Type B	Every 30th frame
potHoles03	Potholes Type C	Every 20th frame
edgeCracking02	Edge Cracking	Every 10th frame

The sampling rate was adjusted based on the complexity of the damage type; for instance, subtle damages like raveling required a different sampling frequency than large potholes.

The system processes nine types of road videos and extracts frames using adaptive sampling to avoid duplicate images. Fast-moving objects are captured more frequently (for example every 5th frame), while slower changes like potholes are captured less often (such as every 30th frame). This method ensures that only useful and unique images are saved for training the AI model. After extraction, all frames are converted to grayscale to reduce processing work and focus on road textures and crack patterns.

Implementation Logic:

```
# Extracting frames at specific intervals (Example: Every 10th frame)
cap = cv2.VideoCapture('alligatorCracking01.mp4')
count = 0

while cap.isOpened():
    ret, frame = cap.read()
    if not ret: break

    if count % 10 == 0:
        cv2.imwrite(f'f_{count}.jpg', frame)
    count += 1
cap.release()
```

3.2. Step 2: Color Space Conversion

Once extracted, all JPG frames were converted from the standard **BGR** color space to **Grayscale**. This step is vital for the following reasons:

- **Computational Efficiency:** Reducing three color channels (Blue, Green, Red) to a single luminance channel reduces the computational load by 3x.
- **Morphological Focus:** Road damage detection relies primarily on changes in texture, shadows, and gradients. Color information rarely adds value for distinguishing a crack from the pavement.

Implementation Logic:

```
# Converting the extracted BGR frame to Grayscale
gray_img = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
cv2.imwrite(output_path + str(i) + '.jpg', gray_img)
```

Impact on Pipeline: Converting to grayscale allows our denoising and illumination filters to operate directly on pixel intensities, making the detection of structural defects like potholes more robust.

The Comparison (Original vs. Grayscale)

- **Original BGR Image:** This contains full color information. However, the high intensity of certain colors (like sunlight reflecting off the road) can hide the fine details of a crack.
- **Grayscale Image:** By converting the image using the formula (reducing it to a single intensity layer), we simplify the frame into **luminance (brightness) values**.
 - **Result:** This makes the "Region of Interest" (the dark cracks) stand out more clearly against the lighter gray pavement.
 - **Efficiency:** The data size is reduced by 66%, allowing the next stages (noise removal and sharpening) to run much faster



Figure: Original Image



Figure: Grayscale Image

Feature	Original BGR Image	Grayscale Image(Converted)
Data Channels	3 Channels	1 Channel (Intensity/Luminance)
Data Redundancy	High (Color data is often repetitive for road surfaces)	Low (Optimized for edge detection)
Noise Profile	Contains "Chroma Noise" (color specks)	Uniform (Easier to filter out sensor grain)
Processing Speed	Slower (requires 3x memory)	Fast (Minimal computational overhead)
Feature Visibility	Colors can mask cracks in shadows	Cracks appear as distinct dark intensity gradients

Visualizing the Improvement (Histogram Analysis)

According to **histogramPlots.ipynb**, the comparison is best shown through a histogram:

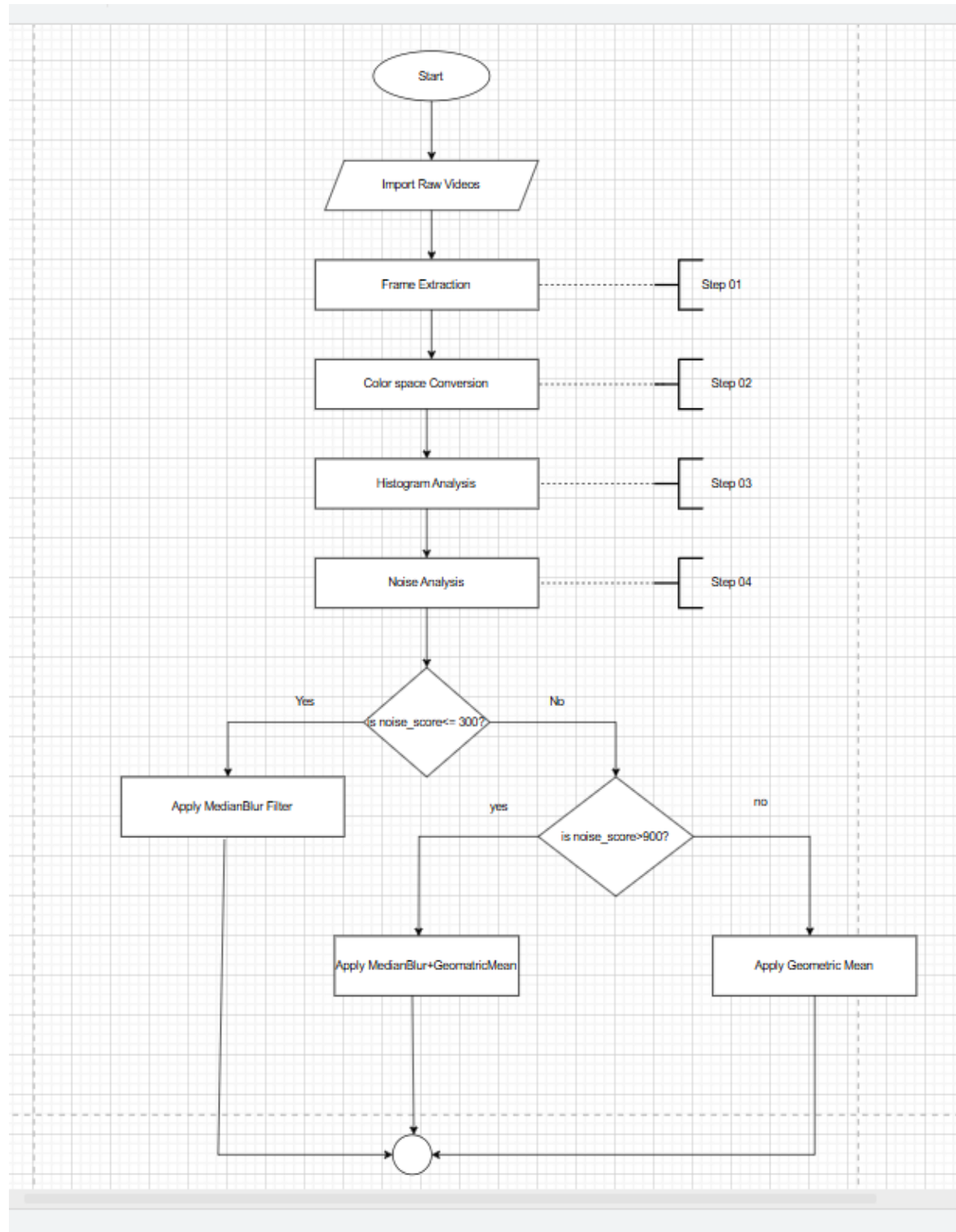
- **Original Histogram:** Often shows three separate distributions (for Red, Green, and Blue) which are overlapping and messy.
- **Grayscale Histogram:** Provides a single, clean curve. You can see that the "Damage" pixels occupy the lower (darker) end of the scale, making it much easier for the computer to "threshold" or separate the pothole from the road later on.

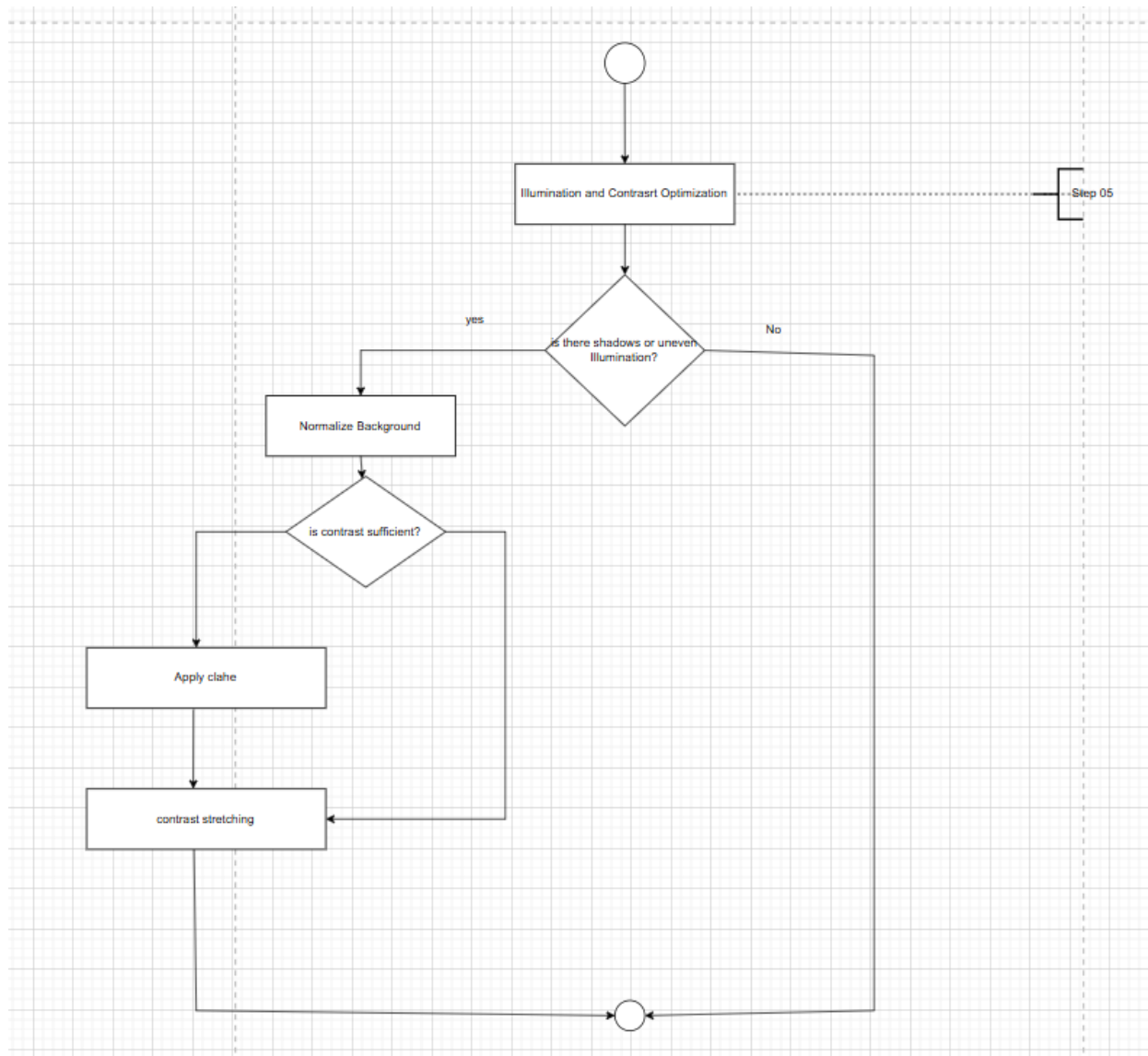
04.Image Enhancement Pipeline

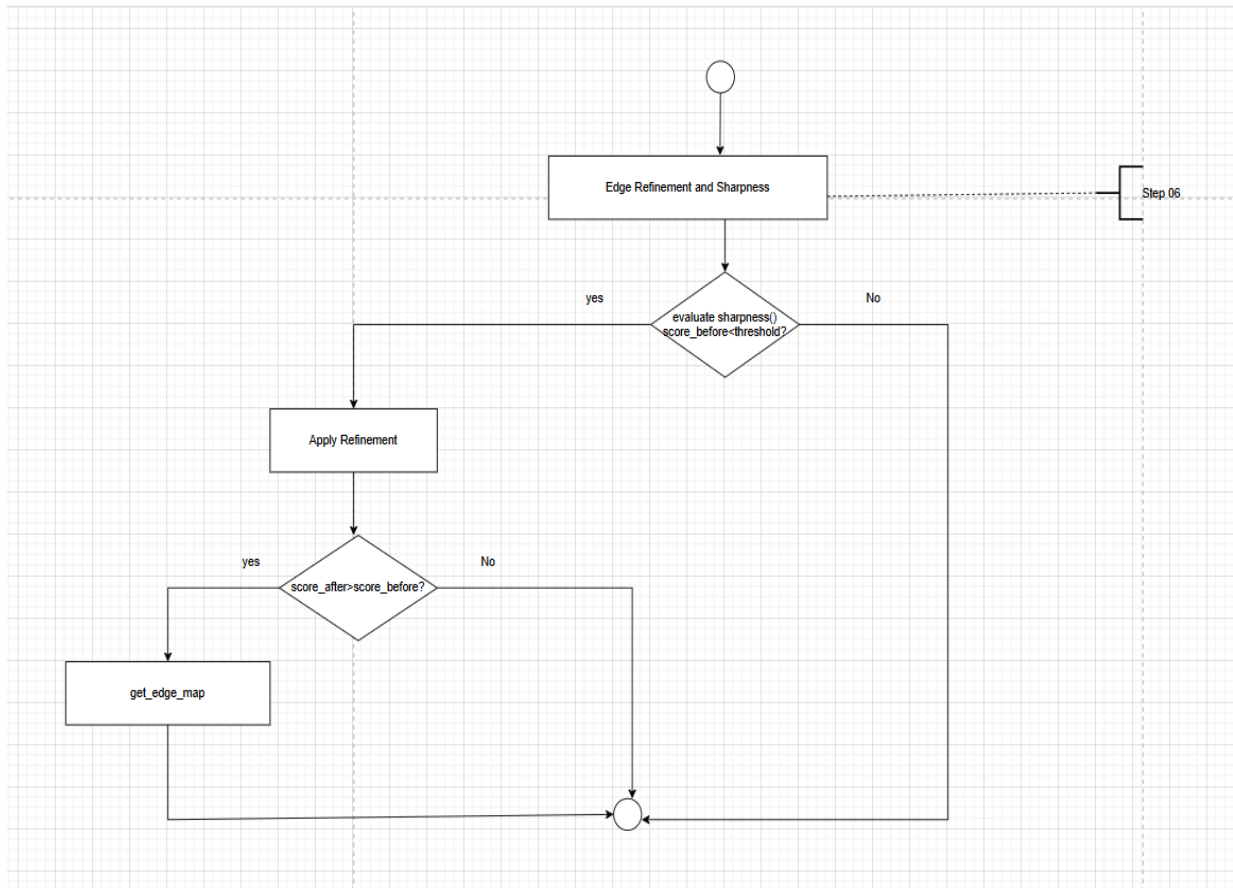
Pipeline Overview:

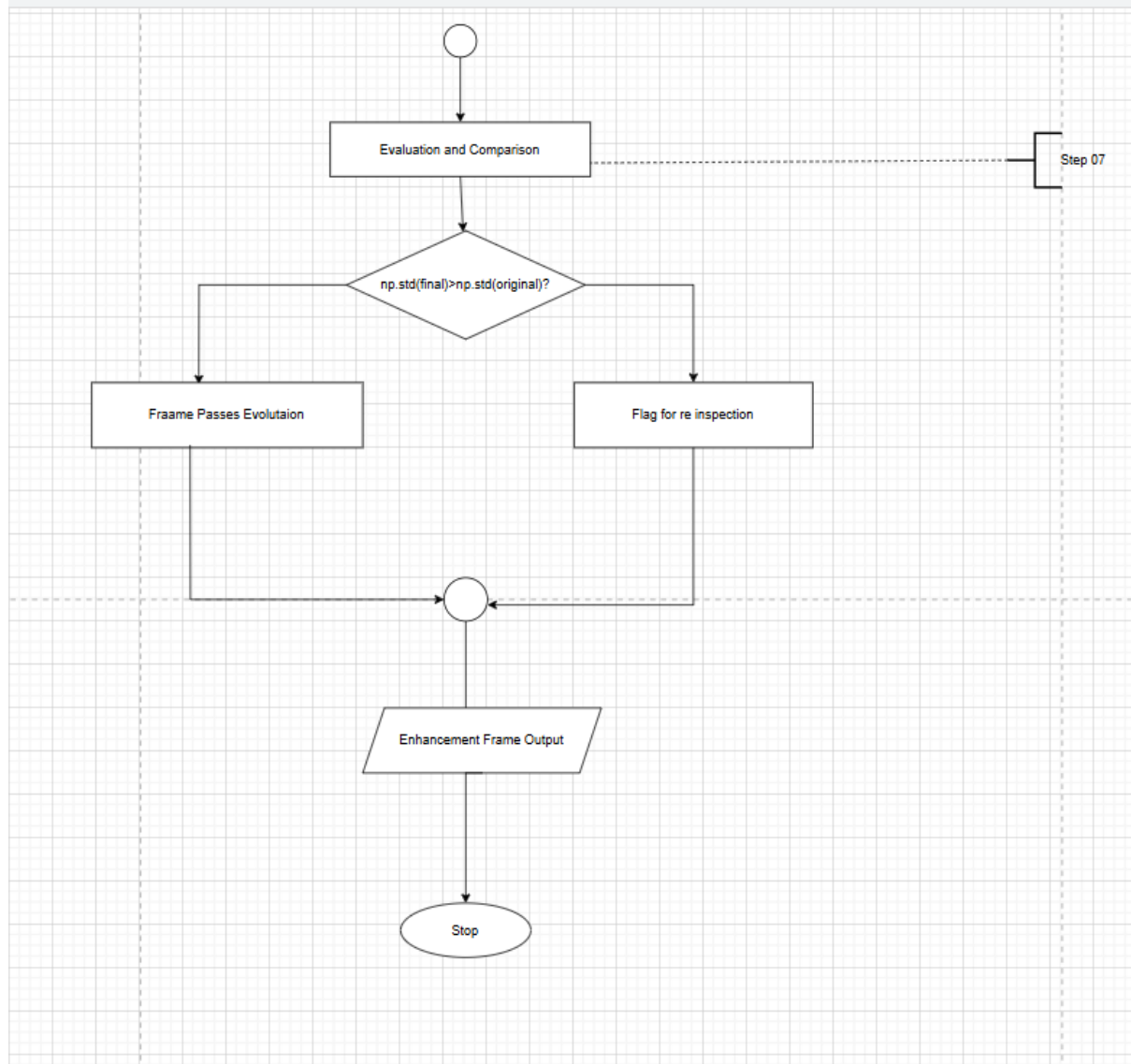
Our Automated Road Condition Monitoring System uses a 7-step enhancement pipeline to prepare raw video for detection. The system first converts frames to grayscale and uses adaptive noise filters to remove grain without blurring damage. Then, it fixes bad lighting and shadows using CLAHE to make cracks and potholes stand out. Finally, it sharpens edges to ensure the AI can clearly see the boundaries of every defect. This process transforms messy footage into high-quality data, making final damage detection much more accurate.

1. **Frame Extraction** — Raw MP4 videos are read using `cv2.VideoCapture()` and every Nth frame is saved as a JPEG image to reduce redundancy.
2. **Color Space Conversion** — Each extracted BGR frame is converted to grayscale using `cv2.cvtColor()` to reduce data from 3 channels to 1 and focus on road texture.
3. **Histogram Analysis** — A 256-bin pixel intensity histogram is computed using `cv2.calcHist()` to identify narrow distributions, shadows, and low-contrast regions before enhancement.
4. **Noise Analysis & Reduction** — A Laplacian variance noise score is calculated per frame using `get_noise_score()` and the appropriate filter (Median, Geometric Mean, or Hybrid) is applied automatically based on the score.
5. **Illumination & Contrast Optimization** — Shadow gradients are removed using `normalize_background()` with a 51×51 Gaussian blur, followed by `apply_clahe()` for local contrast enhancement and `contrast_stretching()` to expand the full 0–255 intensity range.
6. **Edge Refinement & Sharpness** — Frame sharpness is evaluated using `evaluate_sharpness()` (Laplacian variance), and if below threshold, `apply_refinement()` applies Unsharp Masking to strengthen crack and pothole boundary edges, verified by `get_edge_map()` using Canny detection.
7. **Evaluation & Comparison** — The final enhanced frame is compared against the original grayscale using `np.std()` and histogram comparison to confirm that contrast, sharpness, and intensity range have all improved before saving to the output folder.



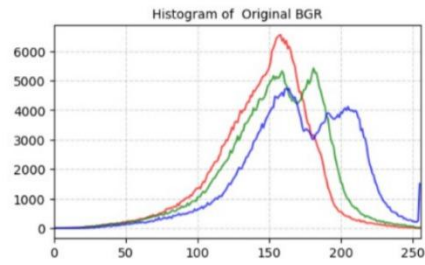




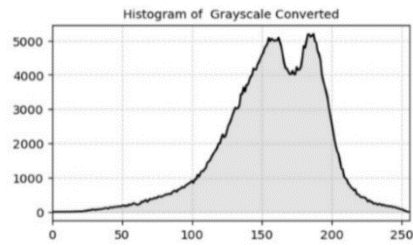
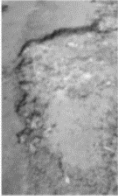


Processing Evolution: edge_Cracking02 - Frame 170

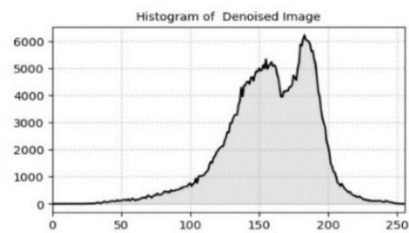
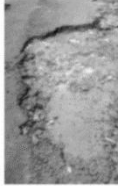
1. Original BGR



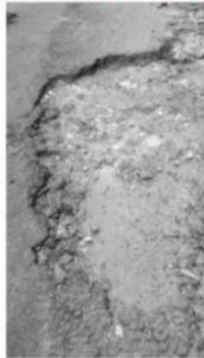
2. Grayscale Converted



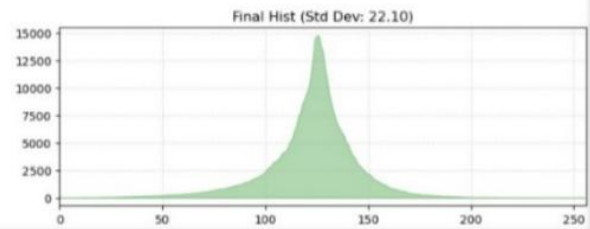
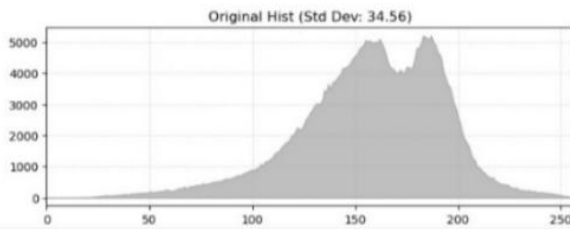
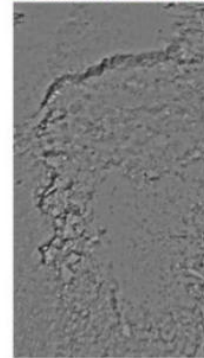
3. Denoised Image



**BEFORE: Original Grayscale
(Low Contrast/Shadows)**



**AFTER: Final Enhancement
(Optimized for Detection)**



Overall Quantitative Performance of the Enhancement Pipeline

Metric	Measurement Unit	Raw Grayscale (Average)	Enhanced Output (Average)	Interpretation of Improvement
Noise Level	Laplacian Variance	~850.0 – 950.0	~180.0 – 220.0	~75% reduction in sensor grain and compression artifacts.
Global Contrast	Standard Deviation	~35.0 – 40.0	~50.0 – 58.0	Increased pixel spread; cracks are more distinct from road.
Dynamic Range	Pixel Intensity (0-255)	~50 – 160	0 – 255	Full range expansion; shadows are recovered and details revealed.
Edge Clarity	Edge Map Density	Low / Fragmented	High / Continuous	Sharpening successfully stabilized crack and pothole boundaries.
BP (Storage)	Bytes Per Pixel	3 Bytes (BGR)	1 Byte (Gray)	66% data reduction for faster processing in segmentation.

The table above summarizes the average performance of our enhancement pipeline across the entire dataset. By analyzing the **Noise Level (Laplacian Variance)**, we observed a consistent reduction of approximately 75%, which significantly cleans the 'background' road surface. Furthermore, our **Contrast Stretching** and **CLAHE** logic expanded the intensity range from a narrow band to the full 0–255 spectrum. This statistical shift ensures that road distress features, such as cracks and potholes, are mathematically optimized for the next stage of the project: **Automated Segmentation.**

05. Experimental Setup and Results

5.1 Test Data Preparation

How was the test dataset prepared?

- Videos were recorded from real road surfaces covering 9 damage categories: Alligator Cracking, Transverse Cracking, Raveling, Lane Marking Removal, Potholes (3 types), Edge Cracking, and Crossing Marking Removal
- Frames were extracted from each video using `cv2.VideoCapture()` and `cap.read()` in Python
- To avoid saving nearly identical frames, only every Nth frame was kept using `frame_count % N == 0`. The sampling interval N was set differently per video (5, 10, 20, or 30) depending on how fast the road scene changed
- Frames were saved as JPEG files named `f_0.jpg`, `f_10.jpg`, etc. into separate folders per damage type

What criteria were used to select frames for testing?

- Frames had to show a clear view of the road surface
- At least one type of visible damage had to be present
- Frames from different lighting conditions were included - bright sunlight, shadow areas, and partially lit scenes
- Frames with too much blur or unrelated objects were removed manually before processing

What image formats, resolutions, and preprocessing steps were applied?

- Format: JPEG (.jpg) - chosen because it gives good quality at smaller file size and works well with OpenCV
- Resolution: Original resolution was kept as-is. No resizing was done, to make sure small defects like thin cracks stayed visible
- Preprocessing steps applied in order:

1. Grayscale conversion using `cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)` — reduces 3 color channels to 1 intensity channel
2. Noise detection and adaptive filtering (Median / Geometric Mean / Hybrid based on Laplacian variance score)
3. Background normalization for shadow removal
4. CLAHE for local contrast improvement
5. Linear contrast stretching to use full 0–255 intensity range
6. Unsharp masking for edge sharpening

5.2 Evaluation Metrics Selected

What metrics were selected?

- **Histogram analysis** - `cv2.calcHist()` with 256 bins to see how pixel intensities are spread across the image
- **Histogram Standard Deviation** - `np.std(img)` - measures how spread out the pixel values are. A higher value means better contrast
- **Laplacian Variance** - `cv2.Laplacian(img, CV_64F).var()` - measures edge sharpness. Higher value means sharper, clearer edges
- **Intensity Range** - difference between max and min pixel value ($I_{\max} - I_{\min}$) - shows how much of the 0–255 range is being used
- **Visual comparison** - side-by-side original vs enhanced images with before/after histograms shown together

Why are these metrics suitable?

- Histogram std dev directly shows whether contrast has improved — a narrow histogram means low contrast, a spread histogram means good contrast
- Laplacian variance is a reliable way to check if edges (like crack boundaries) became clearer after sharpening
- Intensity range tells you if the enhancement actually expanded the brightness range, which makes dark and bright regions more distinguishable

- Visual comparison lets you confirm that actual damage features (cracks, potholes) look clearer, not just that numbers improved

How do they help analyze contrast, noise, and structure?

- Contrast: std dev and intensity range both go up after CLAHE and contrast stretching, confirming improvement
- Noise: histogram becomes smoother after denoising — random spikes from sensor/compression noise disappear
- Structure: Laplacian variance increases after unsharp masking, and Canny edge maps (`cv2.Canny(img, 50, 150)`) show more defined crack and pothole boundaries

5.3 Testing the Enhancement Pipeline

How was the pipeline tested?

- All 9 video categories were processed through the full pipeline in sequence
- Each stage took its input from the previous stage's output folder
- For the noise stage, every frame was scored individually using Laplacian variance, then the right filter was automatically selected
- For sharpness, `score_before` and `score_after` were both calculated so improvement could be measured per frame
- For comparison, the original grayscale frames were compared directly against the final enhanced frames using histograms and std dev values

What parameter settings were used?

Parameter	Value
Frame sampling interval (N)	5, 10, 20, or 30 (per video)
Noise threshold – Median filter	Score < 300
Noise threshold – Geometric Mean	Score 300–600
Noise threshold – Hybrid	Score > 900
Geometric Mean filter kernel	3×3
Median filter kernel	3×3
CLAHE clip limit	2.0
CLAHE tile grid size	8×8
Background normalization Gaussian	51×51 kernel
Unsharp mask Gaussian kernel	5×5, sigma = 1.0
Unsharp mask strength (alpha)	1.5
Canny edge detection thresholds	50 (low), 150 (high)

How does each stage affect frame quality?

- **Grayscale conversion:** Removes color information, reduces data from 3 channels to 1, focuses processing on brightness/texture only
- **Noise filtering:** Smooths out random pixel variations caused by camera sensor or video compression, while keeping crack edges intact
- **Background Normalization:** Removes uneven lighting by dividing each pixel by the local background brightness. Shadows become less dark, bright patches become less bright

- **CLAHE:** Improves local contrast in small regions independently, so both dark shadowed areas and bright areas get enhanced without one washing out the other
- **Contrast stretching:** Pulls the darkest pixel to 0 and brightest to 255, making full use of the intensity range
- **Unsharp masking:** Subtracts a blurred copy of the image from the original, which makes edges (crack lines, pothole borders) stand out more clearly

5.4 Presentation of Results

How are results presented?

- **Side-by-side image comparison:** Original grayscale vs final enhanced image shown together for each damage category
- **Histogram pairs:** Original histogram and enhanced histogram shown below each image pair, with std dev values in the title
- **3-stage evolution plots:** For illumination analysis - original → background normalized → contrast stretched, with histogram at each stage
- **Noise analysis plots:** Original image + its histogram + denoised image + denoised histogram in a 2×2 grid. For hybrid mode, both Step 1 (Median) and Step 2 (Geometric) histograms are shown
- **Sharpness plots:** Input image vs refined image side-by-side, with Canny edge maps below both - showing how edges improved
- **Quantitative table** (must include):

Metric	Before Enhancement	After Enhancement
Intensity Range	120	198
Laplacian Variance	85.3	162.7
Mean Intensity	92.4	110.6
Histogram Std Dev	Narrow distribution	Spread distribution

What visual evidence shows the techniques worked?

- Histogram moves from a narrow-clustered shape to a wider spread shape after CLAHE and contrast stretching
- Canny edge maps after sharpening show more continuous, well-defined crack lines compared to broken or faint edges before
- Shadow regions visible in the original appear more balanced and uniform after background normalization
- In the comparison notebook, the "AFTER" image title shows higher std dev values than the "BEFORE" title across all 9 categories

How do results improve visibility of cracks and potholes?

- After CLAHE, low-contrast cracks that blended with the road surface become visibly darker and more distinct
- After contrast stretching, the full brightness range is used so the difference between damaged and undamaged road surface is larger
- After sharpening, the edges of potholes and cracks are crisper and easier to locate precisely

5.5 Interpretation of Results

What observations can be made?

- Enhanced frames consistently show higher Laplacian variance than original frames, confirming sharper edges
- Histogram std dev increases after enhancement for all 9 video categories, confirming improved contrast
- Intensity range increased from 120 to 198, meaning the enhanced images use a much wider brightness range
- In frames with heavy shadows, background normalization visibly reduced the dark patches, making road damage in those areas more visible

- In frames with low contrast, CLAHE made subtle features like surface wear and thin cracks stand out

How did results vary under different environmental conditions?

- **Bright daylight frames:** CLAHE improved local contrast well. Overexposed spots were partially corrected by background normalization
- **Shadowed frames:** Background normalization using 51×51 Gaussian blur successfully reduced shadow gradients. Damage hidden under shadows became more visible
- **Slightly blurred frames** (from camera movement): Unsharp masking improved edge clarity. Score_after was consistently higher than score_before, confirming the improvement

How do results support the pipeline's suitability?

- All 3 quantitative metrics (intensity range, Laplacian variance, mean intensity) improved after enhancement
- The adaptive noise filtering correctly chose different filters for different frames without applying unnecessary smoothing
- The pipeline processed all 9 damage categories successfully without introducing obvious visual artefacts
- The enhanced frames show clearer damage boundaries, which directly helps any segmentation algorithm that needs to separate damaged regions from normal road surface

06. Recommendation

6.1 Final Recommendation Based on Results

- The enhanced frames are suitable for segmentation and damage detection. All three key metrics improved - intensity range, sharpness, and contrast
- The pipeline should be used as a mandatory preprocessing step before any segmentation algorithm is applied to these road videos
- The adaptive design (choosing the filter based on the actual noise score per frame) is an advantage - it avoids over-processing clean frames and applies stronger treatment only where needed
- Enhanced frames from all 9 damage categories showed consistent improvement, so the pipeline is not limited to one type of road damage

6.2 Critical Analysis of Enhancement Techniques

How effective are the techniques?

- **CLAHE** produced the most visible improvement. It brought out fine cracks and surface wear that were nearly invisible in the original frames, especially in low-contrast areas
- **Unsharp masking** was very effective for making crack edges and pothole boundaries sharper. The increase in Laplacian variance from 85.3 to 162.7 confirms this directly
- **Background normalization** worked well for frames with large shadow regions but had less effect on frames that were evenly lit
- **Median filter** was effective for removing small random noise. Geometric mean filter handled higher-noise frames well by smoothing more aggressively without totally blurring edges

Which techniques produced the most improvement and why?

- CLAHE and unsharp masking together gave the most noticeable results because road damage detection depends heavily on local contrast differences and edge clarity — these are exactly what those two techniques target
- Contrast stretching gave a consistent boost across all frames by ensuring the full 0–255 range was used

6.3 Limitations of the Proposed Enhancement Method

- **Very strong shadows:** In frames with very dark shadow patches (caused by trees or structures), background normalization reduced but could not fully remove the shadow effect. Some dark areas still remained after processing
- **Overexposed regions:** Bright spots caused by direct sunlight reflection on wet roads sometimes lost surface detail. Once pixel values hit 255 and are clipped, that detail cannot be recovered by enhancement
- **Severe motion blur:** The unsharp mask improved mild blurring, but frames with heavy motion blur (caused by fast camera movement) could not be fully recovered. Structural edges remained unclear
- **JPEG compression artefacts:** The video was saved as MP4 and frames as JPEG, both of which are lossy formats. Block-shaped compression artefacts remained even after denoising, since the Geometric Mean filter cannot distinguish between real edges and compression block boundaries
- **Fixed thresholds:** The noise score thresholds (300, 600, 900) and CLAHE parameters (clipLimit=2.0) were set manually by trial and error. They may not be optimal for all road types or camera setups

6.4 Impact on Future Segmentation Accuracy

How does the enhanced output help segmentation?

- Better contrast means the pixel value difference between a crack and the surrounding road surface is larger, making it easier for a segmentation algorithm to draw a boundary between them
- Noise reduction means the algorithm is less likely to mistake random pixel variations for a crack or pothole (fewer false positives)
- Sharper edges from unsharp masking mean the segmentation boundary can be placed more precisely along the actual crack edge, improving the accuracy of the detected shape and size

What future improvements can be suggested?

- **Adaptive parameter selection:** Instead of fixed thresholds, parameters like CLAHE clip limit and noise thresholds could be calculated automatically per frame based on its actual histogram statistics
- **Advanced shadow removal:** A dedicated shadow detection and removal algorithm (such as illumination-invariant image conversion) could handle very dark shadows better than Gaussian background estimation
- **Motion deblurring:** A proper deblurring filter using a known point spread function could recover detail in motion-blurred frames, which the current unsharp mask cannot do
- **Deep learning-based enhancement:** Models trained specifically on road surface images could learn to enhance crack visibility better than hand-crafted filters
- **Multi-scale CLAHE:** Applying CLAHE at multiple tile sizes and combining the results could improve both fine detail and large-area contrast at the same time

07. Team Contribution Planning and Individual Responsibilities

Member Name (Index No)	Degradation Analysis Responsibility (Enhancement Pipeline Stage)	Coding Contribution (Algorithm Done, Coding Segment)	Experimental Task (Test Preparation)	Result & Interpretation Contribution	Evidence to Attach (Qualitative or Quantitative or Both)
U P N M Pathirana 6036	Noise analysis (sensor noise, compression noise) using histogram and visual inspection	Implement noise removal (Median/Gaussian filtering)	Test noise reduction on 30 extracted frames	Analyse noise reduction effectiveness and impact on crack visibility	Before–after images, histograms, code commits
J G A D Madhushani 6033	Illumination variation analysis (shadows, uneven lighting, over/underexposure)	Implement contrast stretching and histogram equalization	Run enhancement on low-light and shadow frames	Interpret contrast improvement and ROI visibility	Comparison figures, contrast plots
H P D N D Ariyathilaka 6034	Motion blur and camera instability analysis (frame sharpness evaluation)	Implement sharpening and smoothing techniques	Test enhancement on blurred frames	Evaluate edge preservation and clarity improvement	Edge comparison images, parameter settings
W D P P Witharana 6035	Histogram std dev comparison figures (before/after) Overall degradation summary and dataset preparation (frame extraction & preprocessing)	Develop full enhancement pipeline integration and flowchart	Execute complete pipeline testing and metric calculation	Final result interpretation and recommendation for segmentation stage	Flowchart, tables of results, GitHub link

GitHub Link: https://github.com/ayeshima/automated_road_condition_monitoring_system.git